WEEK 2 · LESSON 11 OF 24 · HANDOUT

Charts with Matplotlib

Study time: **about 45 minutes** (reading, video and activity)

▶ **Watch:** https://www.youtube.com/watch?v=a9UrKTVEeZA — open the lesson online to watch inside the course.

🎓 **Go deeper:** freeCodeCamp: Data Analysis with Python (NumPy, Pandas, Matplotlib) — full course — https://www.youtube.com/watch?v=r-uOLxNrNk8

Your goal in this lesson: Choose the right chart for a question, draw line, bar, histogram and scatter charts with Matplotlib, and label them so anyone can read them.

Pick the chart from the question

Question	Chart
How does a value change over time?	**Line** chart
How do categories compare?	**Bar** chart
How are values spread out?	**Histogram**
Are two numbers related?	**Scatter** plot

Pie charts are hard to read when there are more than 2–3 slices; a bar chart is almost always clearer.

Run the setup cell first:

```
import numpy as np
import pandas as pd

rng = np.random.default_rng(7)
n = 200
prices = {"Rice 5kg": 7500, "Cooking oil 1L": 3200, "Sugar 1kg": 1600,
          "Milk 1L": 800, "Airtime 1000": 1000}
products = rng.choice(list(prices), size=n, p=[0.2, 0.15, 0.2, 0.3, 0.15])
sales = pd.DataFrame({
    "date": pd.to_datetime("2026-01-01") + pd.to_timedelta(rng.integers(0, 181, n), unit="D"),
    "district": rng.choice(["Gasabo", "Kicukiro", "Nyarugenge"], size=n),
    "product": products,
    "quantity": rng.integers(1, 7, n),
    "payment": rng.choice(["MoMo", "Cash", "Card"], size=n, p=[0.55, 0.35, 0.10]),
})
sales["unit_price"] = sales["product"].map(prices)
sales["total"] = sales["quantity"] * sales["unit_price"]
sales = sales.sort_values("date").reset_index(drop=True)
```

```
import matplotlib.pyplot as plt
```

Line chart — change over time

```
monthly = sales.groupby(sales["date"].dt.month)["total"].sum()

plt.figure(figsize=(8, 4))
plt.plot(monthly.index, monthly.values, marker="o")
plt.title("Monthly income, January–June 2026")
plt.xlabel("Month")
plt.ylabel("Income (RWF)")
plt.grid(alpha=0.3)
plt.show()
```

Bar chart — compare categories

```
by_product = sales.groupby("product")["total"].sum().sort_values()

plt.figure(figsize=(8, 4))
plt.barh(by_product.index, by_product.values, color="#1f8a9e")
plt.title("Income by product")
plt.xlabel("Income (RWF)")
plt.tight_layout()
plt.show()
```

Horizontal bars (`barh`) make long labels easy to read. Sorting the bars makes the ranking obvious.

Histogram — the spread of values

```
plt.figure(figsize=(8, 4))
plt.hist(sales["total"], bins=20, color="#7c5cbf", edgecolor="white")
plt.title("How big are our sales?")
plt.xlabel("Sale total (RWF)")
plt.ylabel("Number of sales")
plt.show()
```

A histogram puts values into **bins** and counts how many fall in each. It shows whether most sales are small, whether there are a few very big ones, and where the middle is.

Scatter plot — relationships

```
plt.figure(figsize=(6, 4))
plt.scatter(sales["quantity"], sales["total"], alpha=0.5)
plt.title("Quantity vs sale total")
plt.xlabel("Quantity")
plt.ylabel("Total (RWF)")
plt.show()
```

Scatter plots are how you first spot patterns that a model could learn (Week 4).

Several charts together

```
fig, axes = plt.subplots(1, 2, figsize=(10, 4))
sales["payment"].value_counts().plot(kind="bar", ax=axes[0], title="Sales by payment")
sales.groupby("district")["total"].mean().plot(kind="bar", ax=axes[1], title="Average sale by
district")
plt.tight_layout()
plt.show()
```

Pandas columns have a built-in `.plot()` that uses Matplotlib underneath.

Save a chart

```
plt.figure(figsize=(8, 4))
plt.plot(monthly.index, monthly.values, marker="o")
plt.title("Monthly income")
plt.savefig("monthly_income.png", dpi=150, bbox_inches="tight")
```

Download it from the Colab folder to use in a report or slide.

Rules for honest, clear charts

1. A **title** that says what the chart shows.
2. **Axis labels with units** (RWF, %, km).
3. Bar charts start at **zero** — a cut axis exaggerates differences.
4. One message per chart.
5. Mention the data source and period.

Key terms

- **Line / bar / histogram / scatter** — the four core chart types.
- **Bins** — the ranges a histogram counts values into.
- **Figure / axes** — the whole picture / one chart inside it.
- **subplots** — several charts in one figure.
- **savefig** — save a chart as an image file.

Remember this: Choose the chart from the question: line for time, bar for comparison, histogram for spread, scatter for relationships. Always add a title, axis labels and units.

Name: Date:

Activity

Task 1

Draw a bar chart of the number of sales per district.

Task 2

Draw a line chart of monthly MoMo income and monthly Cash income on the same axes (call `plt.plot` twice and add `plt.legend()`).

Task 3

Draw a histogram of `quantity`. What is the most common quantity?

Task 4

Save your best chart as a PNG and write a one-sentence caption for it.

Self-check

Think about your answers here, then take the quiz online — it is marked instantly and explains every answer.

1. Which chart shows how income changes month by month?
   - A. Line chart
   - B. Scatter plot
   - C. Pie chart
   - D. Histogram
2. Which chart shows how sale amounts are spread out?
   - A. Line chart
   - B. Histogram
   - C. Pie chart
   - D. Table
3. Which chart helps you see if two numbers are related?
   - A. Histogram
   - B. Scatter plot
   - C. Pie chart
   - D. Bar chart
4. Why should a bar chart start at zero?
   - A. It looks nicer
   - B. Cutting the axis exaggerates differences and can mislead
   - C. Matplotlib needs it
   - D. It does not matter
5. What does `plt.savefig("chart.png")` do?
   - A. Prints the data
   - B. Deletes the chart
   - C. Changes colours
   - D. Saves the chart as an image file
6. Which is NOT one of the rules for a clear chart?
   - A. Mention the data source
   - B. As many messages as possible in one chart
   - C. A title
   - D. Axis labels with units

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